Intermediate Macroeconomics

Tutorial 2 Consumption

Answer

1. This lecture analyzes Fisher's model for the case in which the consumer can save or borrow at an interest rate of r and for the case in which the consumer can save at this rate but cannot borrow at all. Consider now the intermediate case in which the consumer can save at rate r_s and borrow at rate r_b, where $r_s < r_b$.

   (a) What is the consumer's budget constraint in the case in which he consumes less than his income in period one? Answer in the form of an equation.

   Answer:

   A consumer who consumes less than his income in period one is a saver and faces an interest rate r_s. His budget constraint is

   $$C_1 + \frac{C_2}{1+r_s} = Y_1 + \frac{Y_2}{1+r_s}.$$

   (b) What is the consumer's budget constraint in the case in which he consumes more than his income in period one? Answer in the form of an equation.

   Answer:

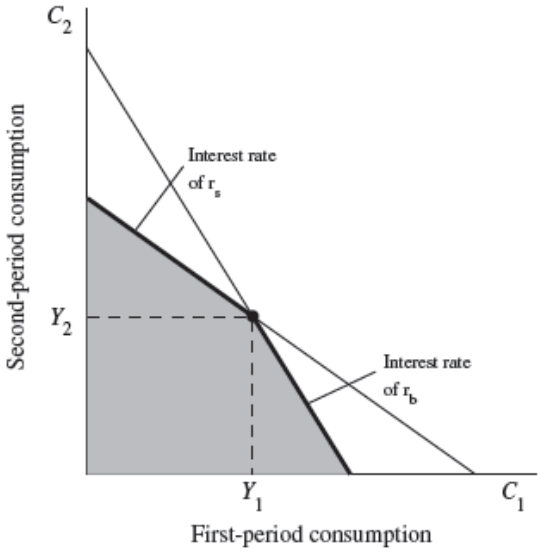
   A consumer who consumes more than income in period one is a borrower and faces an interest rate r_b. The budget constraint is

   $$C_1 + \frac{C_2}{1+r_b} = Y_1 + \frac{Y_2}{1+r_b}.$$

   (c) On a single graph, show the two budget constraints from parts (a) and (b). Shade the area that represents the combination of first-period and second-period consumption the consumer can choose.

   Answer:

   The following figure shows the two budget constraints; they intersect at the point (Y_1, Y_2), where the consumer is neither a borrower nor a lender. The shaded area represents the combinations of first-period and second-period consumption that the consumer can choose. To the left of the point (Y_1, Y_2), the interest rate is r_b.



Budget line with constraints

(d) Now add to your graph the consumer's indifference curves. Show three possible outcomes: one in which the consumer saves, one in which he borrows, and one in which he neither saves nor borrows.

Answer:

Figure 2 shows the three cases. Figure 2(A) shows the case of a saver for whom the indifference curve is tangent to the budget constraint along the line segment to the left of (Y_1, Y_2). Figure 2(B) shows the case of a borrower for whom the indifference curve is tangent to the budget constraint along the line segment to the right of (Y_1, Y_2). Finally, Figure 2(C) shows the case in which the consumer is neither a borrower nor a lender: the highest indifference curve the consumer can reach is the one that passes through the point (Y_1, Y_2).

(e) What determines first-period consumption in each of the three cases?

Answer:

If the consumer is a saver, then consumption in the first period depends on $[Y1 + Y2/(1 + r_s)]$—that is, income in both periods, Y_1 and Y_2, and the interest rate r_s. If the consumer is a borrower, then consumption in the first period depends on $[Y_1 + Y_2/(1 + r_b)]$—that is, interest rate r_b. Note that borrowers discount future income more than savers.

If income in both periods, Y_1 and Y_2, and the the consumer is neither a borrower nor a lender, then consumption in the first period depends just on Y_1.

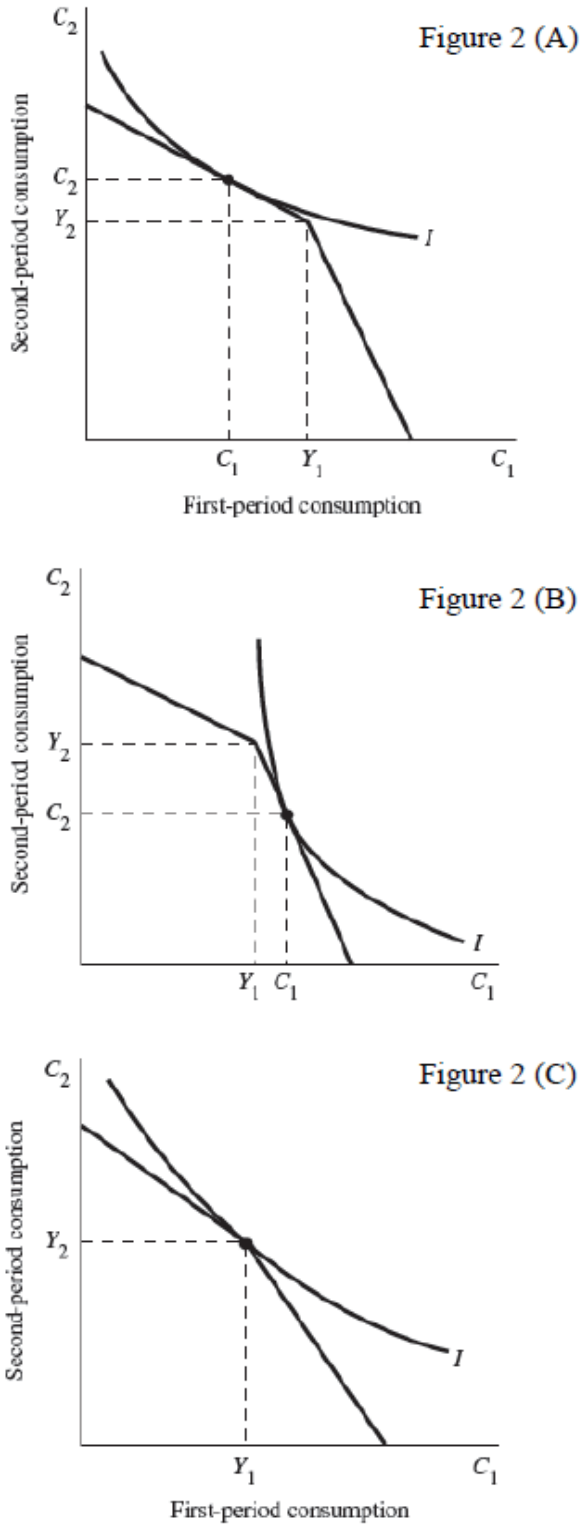


Figure 1: Consumption patterns with budget constraints

2. A representative consumer lives for two periods and has the following utility function

$$U(C_1, C_2) = \ln C_1 + \beta \ln C_2,$$

where $0 < \beta < 1$ is the exogenous discount factor. The present-value budget constraint faced is given by

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}.$$

where r is the exogenous real rate of interest, and both first-period income Y_1 and second-period income Y_2 are exogenously given.

(a) Set up the Lagrangian function and derive the first-order conditions.

Answer:

The Lagrangion function:

$$\mathcal{L} = \ln C_1 + \beta \ln C_2 + \lambda \left[Y_1 + \frac{Y_2}{1+r} - C_1 - \frac{C_2}{1+r} \right],$$

where λ is the Lagrange multiplier.

The three first-order conditions for C_1, C_1, and λ are

$$\frac{\partial \mathcal{L}}{\partial C_1} = 0 : \frac{1}{C_1} = \lambda, \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial C_2} = 0 : \frac{\beta}{C_2} = \frac{\lambda}{1+r}, \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 : C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}. \tag{3}$$

(b) Solve for the optimal C_1 and C_2.

Answer:

From (1) and (2) we have

$$C_2 = (1+r)\beta C_1. \tag{4}$$

Substitute (4) into (3) and rearrange, we obtain the optimal C_1

$$C_1 = \frac{1}{1+\beta} \left[Y_1 + \frac{Y_2}{1+r} \right]. \tag{5}$$

Substitute (5) back to (4), we obtain the optimal C_2

$$C_2 = \frac{\beta(1+r)}{1+\beta} \left[Y_1 + \frac{Y_2}{1+r} \right].$$

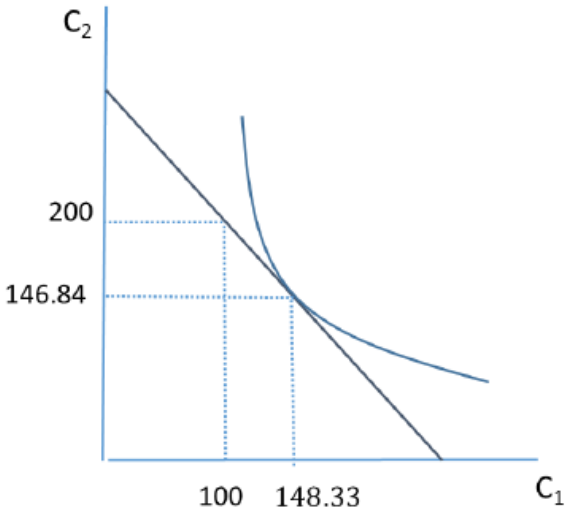
(c) If a person has $\beta = 0.9$, $Y_1 = 100$, and $Y_2 = 200$ and the interest rate is 10%. Calculate the consumption levels in both periods. Draw a graph to show his consumption allocation. Is he a saver or a borrower?

Answer:

Substitute above figures into answers in (b), we have

$$\begin{aligned} C_1 &\approx 148.33, \\ C_2 &\approx 146.84. \end{aligned}$$

He is a borrower because he consumes more than what he has in the first period.



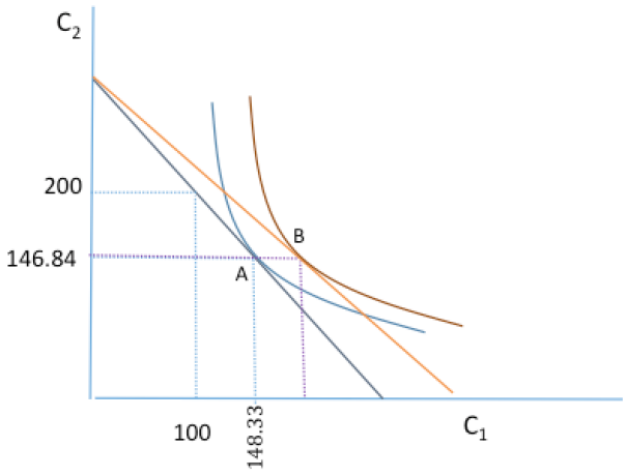
(d) If this person earns nothing in the first period but earns an amount equal to $(1 + 10\%)100 + 200$ in the second period. Does that change your answers in (a), (b), and (c)?

Answer:

No, because his intertemporal budget line is unchanged. he has just need to borrow more to consume in period 1.

(e) Now still in the context of (d), what happens to the budget line when the interest rate falls? Which effect dominates in this case – income or substitution

Answer:



Note that consumer is a borrower in period 1. The fall of interest rate does not change the consumer's endowment in period 2 and the budget line will **pivot around the endowment point** $(0, (1+10\%)100+200)$.

Substitution effect causes C_1 to increase while C_2 to decrease. Because C_1 can be financed with lower costs due to the fall in interest rate. The consumer tends to substitute C_2 for the relatively cheaper C_1.

Meanwhile, the **income effect** implies that both C_1 and C_2 will rise. A fall in interest rate increases the consumer's real income so that he can affort more of both C_1 and C_2.

Both substitution and income effects increase C_1 but lead to ambiguous effects on C_2. In fact, substitution effect and income effect just offset each other, as you will see from the following derivation.

Given the endowment points $(0, 310)$, we can compute C_1 and C_2 from part (b) as:

$$\begin{aligned} C_1 &= \frac{310}{(1+\beta)(1+r)}, \\ C_2 &= \frac{\beta \times 310}{1+\beta}. \end{aligned}$$

As we can see, a drop in interest rate r will cause C_1 to increase and no effect on C_2.

3. **Life-cycle saving.** (Modigliani and Brumberg, 1954)

Consider an individual who lives from 0 to T, and whose lifetime utility is given by $U = \int_{t=0}^{T} u(C(t))\,dt$, where $u'(\cdot) > 0$, $u''(\cdot) < 0$. The individual's income is $Y_0 + gt$ for $0 < t < R$, and 0 for $R \leq t \leq T$. The retirement age, R, satisfies $0 < R < T$. The interest rate is zero, the individual has no initial wealth, and there is no uncertainty.

(a) What is the individual's lifetime budget constraint?

Answer:

The present value of lifetime consumption must be less than or equal to the present value of lifetime income (the individual has no initial wealth). Thus we have

$$\int_{t=0}^{T} C(t)dt \leq \int_{t=0}^{T} Y(t)dt. \tag{6}$$

Since the individual's income is

$$Y(t) = \begin{cases} Y_0 + gt & \text{for } 0 < t < R \\ 0 & \text{for } R \leq t \leq T \end{cases}$$

The present value of lifetime income is

$$\int_{t=0}^{T} Y(t)dt = \int_{t=0}^{R} (Y_0 + gt)\,dt.$$

Solving this integral gives us

$$\begin{aligned}\int_{t=0}^{R}(Y_0+gt)\,dt &= \left[Y_0t+\frac{1}{2}gt^2\right]\Big|_{t=0}^{R}\\ &= RY_0+\frac{1}{2}gR^2. \qquad (7)\end{aligned}$$

Substituting equation (7) into equation (6) gives us the lifetime budget constraint

$$\int_{t=0}^{T}C(t)dt \leq RY_0+\frac{1}{2}gR^2.$$

(b) What is the individual's utility-maximizing path of consumption, $C(t)$?

Answer:

Since $u''(\cdot) < 0$ and the interest rate and discount rate are equal to zero here, the analysis in the Permanent Income Hypothesis (PIH) implies that utility-maximization requires consumption to be constant. The budget constraint then implies that consumption at each point in time is equal to lifetime resources divided by the length of life. As shown in part (a), lifetime resources are given by $RY_0+\frac{1}{2}gR^2$ and thus the constant level of consumption is given by

$$\overline{C}=\frac{R}{T}\left[Y_0+\frac{1}{2}gR\right]. \qquad (8)$$

(c) What is the path of the individual's wealth as a function of t? (Optional)

Answer:

$$W(t)=\int_{\tau=0}^{t}S(\tau)d\tau,$$

where S represents the individual's saving and W represents her wealth. Saving at time t is the difference between income and consumption, or

$$S(t)=Y(t)-C(t).$$

Thus saving is given by

$$S(t)=\begin{cases}Y_0+gt-\overline{C} & \text{for } 0<t<R\\ -\overline{C} & \text{for } R\leq t\leq T\end{cases}$$

Intuitively, the individual dissaves when current income is less than mean lifetime income (as it is in retirement when current income is zero) and saves when current income is greater than mean lifetime income.

For $0<t<R$, wealth is given by

$$W(t)=\int_{\tau=0}^{t}S(\tau)d\tau=\int_{\tau=0}^{t}\left[Y_0+g\tau-\overline{C}\right]d\tau.$$

Solving the integral gives us

$$\begin{aligned} W(t) &= \left[Y_0\tau + \frac{1}{2}g\tau^2 - \overline{C}\tau \right] \Big|_{\tau=0}^{t} \\ &= Y_0 t + \frac{1}{2}gt^2 - \overline{C}t \\ &= t\left(Y_0 + \frac{1}{2}gt - \overline{C} \right). \end{aligned} \tag{9}$$

For $R \leq t \leq T$, wealth is given by

$$W(t) = \int_{\tau=R}^{t} S(\tau)d\tau + W(R), \tag{10}$$

where $W(R)$ is wealth at retirement which can be found by substituting $t = R$ into (9),

$$W(R) = R\left(Y_0 + \frac{1}{2}gR - \overline{C} \right). \tag{11}$$

Since $\overline{C} = \frac{R}{T}\left[Y_0 + \frac{1}{2}gR \right]$, equation (11) can be further writen as

$$\begin{aligned} W(R) &= R\left(\frac{T}{R}\overline{C} - \overline{C} \right) \\ &= (T - R)\,\overline{C}. \end{aligned} \tag{12}$$

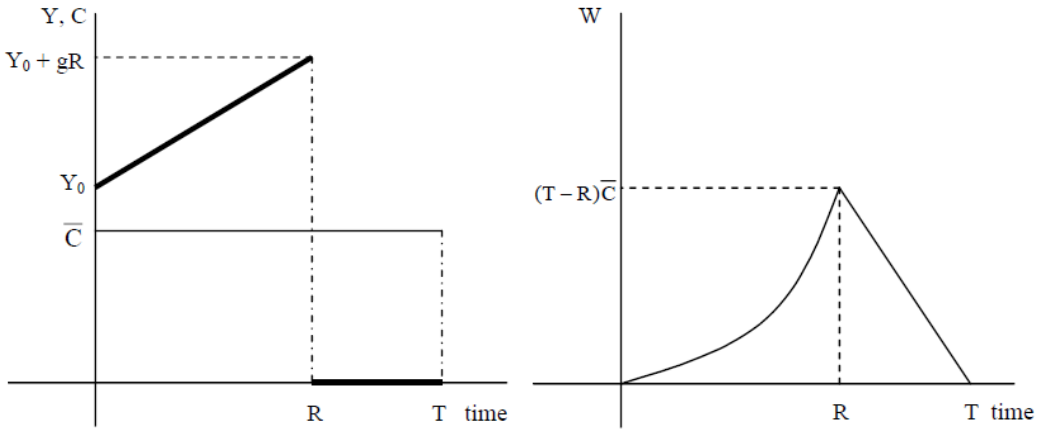
Equation (12) is intuitive: since the individual receives no income in retirement, consumption when retired must be financed by the wealth built up during the individual's working life. Since $(T - R)$ is the amount of time spent in retirement, and since the individual consumes $\overline{C}$ per unit time, wealth at retirement must equal $(T - R)\,\overline{C}$.

Substituting equation (12) and the fact that $S(t) = -\overline{C}$ for $R \leq t \leq T$ into equation (10) yields

$$\begin{aligned} W(t) &= (T - R)\,\overline{C} - \int_{\tau=R}^{t} \overline{C}d\tau \\ &= (T - R)\,\overline{C} - (t - R)\overline{C} \\ &= (T - t)\,\overline{C}. \end{aligned} \tag{13}$$

Thus the individual begins positive saving once current income exceeds mean lifetime income. Wealth is maximized at retirement, after which wealth is run down to finance consumption until – from equation (13) – wealth equals zero at time T, the end of life.

The pattern implied by our analysis is shown in the figures below.



The left-hand figure depicts income and consumption as functions of time, assuming that income at time 0 exceeds the constant level of consumption. The bold line shows income which equals $Y_0 + gt$ until retirement and equals 0 thereafter. Consumption is constant at the level $\overline{C}$.

The right-hand figure depicts wealth as a function of time. The slope of the wealth curve equals saving which in turn is equal to the difference between income and consumption in the figure on the left. Wealth rises (at an increasing rate) during working life since income exceeds consumption (by an increasing amount); wealth reaches a maximum of $(T - R)\,\overline{C}$ at retirement. During retirement – between periods R and T – wealth declines at a constant rate until it reaches zero at the end of life. Given the shape of the wealth function, this pattern of wealth accumulation over the life cycle is known as "hump saving".